

6(a)	- Electrons in the gas atoms interact with the photons and the photon energy causes the electron to move to higher energy level/to be excited - Since the energy of the electron energy level is discrete, the difference in the energy will be discrete - Only photon energy equivalent to the difference of energy of the electron energy levels will cause this excitation - When the electrons de-excite, photons are emitted in all directions and hence they appear to be dark lines [point 1 and point 3 are compulsory], [3 out of 4 points]	1 1 1 1	
(b)(i)	6	1	
(ii)1.	Maximum photon energy = $13.6 - 0.85$ = 12.75 Maximum KE= $12.75 - 5.6$ = 7.2 (7.15) eV	1 1	
2.	Wavelength = $(6.63 \times 10^{-34})(3 \times 10^8) / (12.75 \times 1.6 \times 10^{-19})$ = 9.75×10^{-8} m	1	
3.	Velocity of electron = $\sqrt{\frac{2(7.2 \times 1.6 \times 10^{-19})}{9.11 \times 10^{-31}}} = 1.59 \times 10^6$ Uncertainty of momentum = $(9.11 \times 10^{-31})(0.15 \times 1.59 \times 10^6) = 2.17 \times 10^{-25}$ Uncertainty of position = $(6.63 \times 10^{-34}) / (2.17 \times 10^{-25}) = 3.06 \times 10^{-9}$ m	1 1 1	
(c)(i)	Either: If waves (or light) passes through suitable aperture size, Diffraction would occur and similar pattern would be observed Or: Concentric circles are evidence of diffraction And diffraction is a wave property	1 1 1 1	
(ii)	(speed increases so) momentum increases Since $\lambda = h/p$ so λ decreases, hence radii decreases	1 1	